

WEEK 2 REPORT

Data Modeling, DAX & Dashboard Creation using Power BI

Introduction

In Week 2, I worked on transforming the cleaned dataset into meaningful insights using **Power BI**. The focus of this week was on **data modeling, creating DAX measures, and designing an interactive dashboard**. This process helped in understanding how raw data can be converted into useful business information.

Step 1: Loading the Cleaned Data

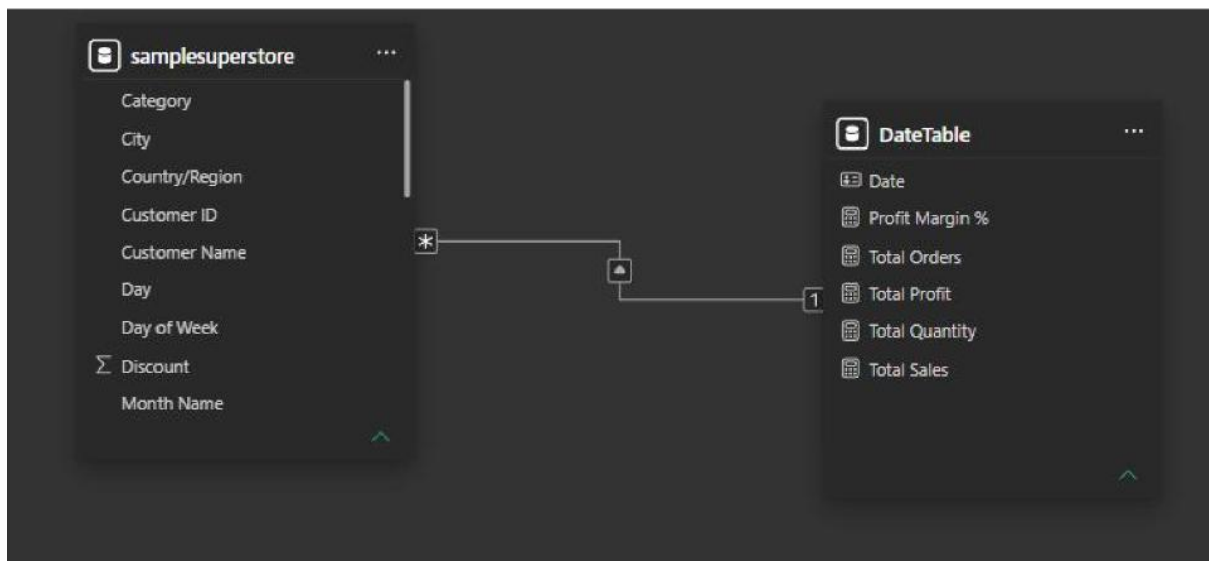
After completing the data cleaning process in Power Query during Week 1, the dataset was loaded into Power BI by clicking the **“Close & Apply”** option. This step applies all the transformations and prepares the data for analysis.

= Table.TransformColumns(#"Changed Type with Locale1",{{"Product Name", Text.Upper, type text}, {"Category", Text.Upper, type text}}										
	A ^B _C Ship Mode	A ^B _C Customer ID	A ^B _C Customer Name	A ^B _C Segment	A ^B _C Country/Region	A ^B _C City				
100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%
7-01-2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	HOUSTON				
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL				
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL				
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL				
2-01-2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	PHILADEL				
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK				
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK				
7-01-2023	First Class	JO-15145	Jack O'Briant	Corporate	United States	ATHENS				
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK				
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK				
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK				
8-01-2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	LOS ANGE				
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK				
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK				
2-01-2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	HUNTSVIL				
2-01-2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	HUNTSVIL				
3-01-2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	LAREDO				
3-01-2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	LAREDO				
5-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	SPRINGFIE				
5-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	SPRINGFIE				
4-01-2023	First Class	SV-20365	Seth Vernon	Consumer	United States	DOVER				
8-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States	SAN FRAN				
6-01-2023	Second Class	ND-18370	Natalie DeCherney	Consumer	United States	MOUNT PI				
8-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States	SAN FRAN				
0-01-2023	Standard Class	MM-17320	Michael Moore	Consumer	United States	MEMPHIS				

Step 2: Data Modeling

The next step was to check the structure of the data using the **Model View** in Power BI.

The dataset used (**Sample Superstore**) contains only a **single table**, so there was no need to create relationships between tables. Therefore, the data modeling step was simple, and I proceeded directly to creating calculations.



Step 3: Creating DAX Measures

To analyze the data effectively, I created several **DAX (Data Analysis Expressions) measures**. These measures help in calculating key performance indicators (KPIs).

Measures Created:

1. Total Sales

Total Sales = SUM(samplesuperstore[Sales])

2. Total Profit

Total Profit = SUM(samplesuperstore[Profit])

3. Total Orders

Total Orders = COUNT(samplesuperstore[Order ID])

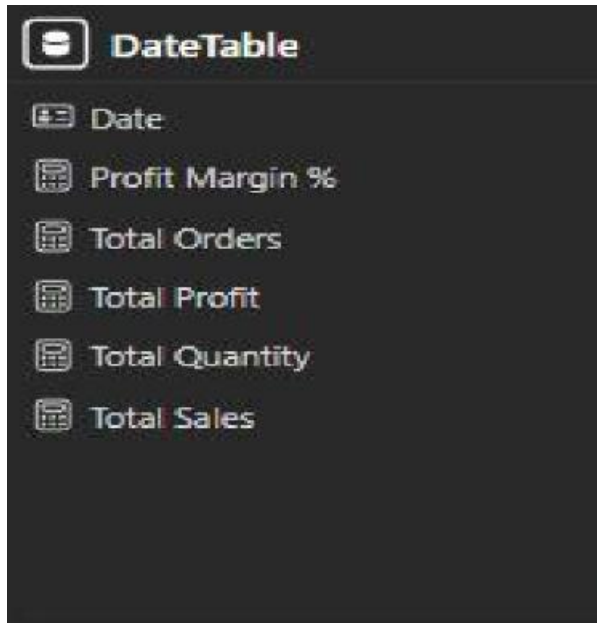
4. Total Quantity

Total Quantity = SUM(samplesuperstore[Quantity])

5. Profit Margin %

Profit Margin % = $\text{DIVIDE}([\text{Total Profit}], [\text{Total Sales}], 0) * 100$

These measures were used to evaluate overall business performance.



Step 4: Dashboard Creation

An interactive dashboard was created using different types of visuals to represent the data clearly.

KPI Cards:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin %

These cards provide a quick summary of key metrics.

2.33M

Total Sales

Tot

Total Sales

10K

Total Orders

292.30K

Total Profit

12.56

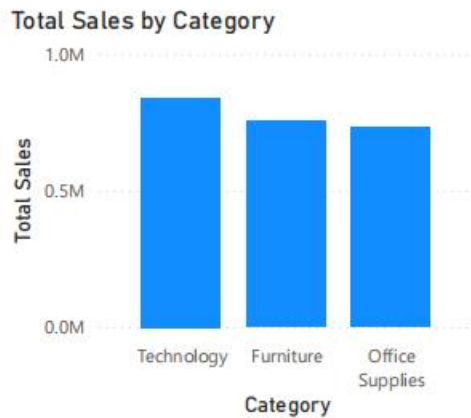
Profit Margin %



Bar Chart:

- **Axis:** Category
- **Values:** Total Sales

This chart shows the comparison of sales across different product categories.



Line Chart:

- **Axis:** Order Date
- **Values:** Total Sales

This chart helps in understanding the sales trend over time.



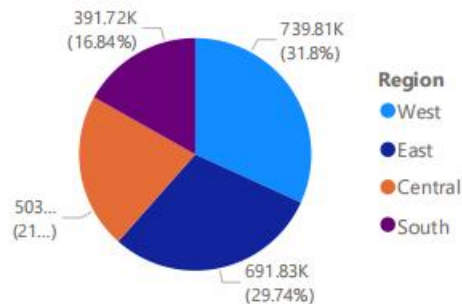
Pie Chart:

- **Legend:** Region

- **Values:** Total Sales

This chart represents the contribution of each region to total sales.

Total Sales by Region



Slicers (Filters):

- Region
- Category
- Segment
- Order Date

Slicers were added to make the dashboard interactive, allowing users to filter and explore data dynamically.



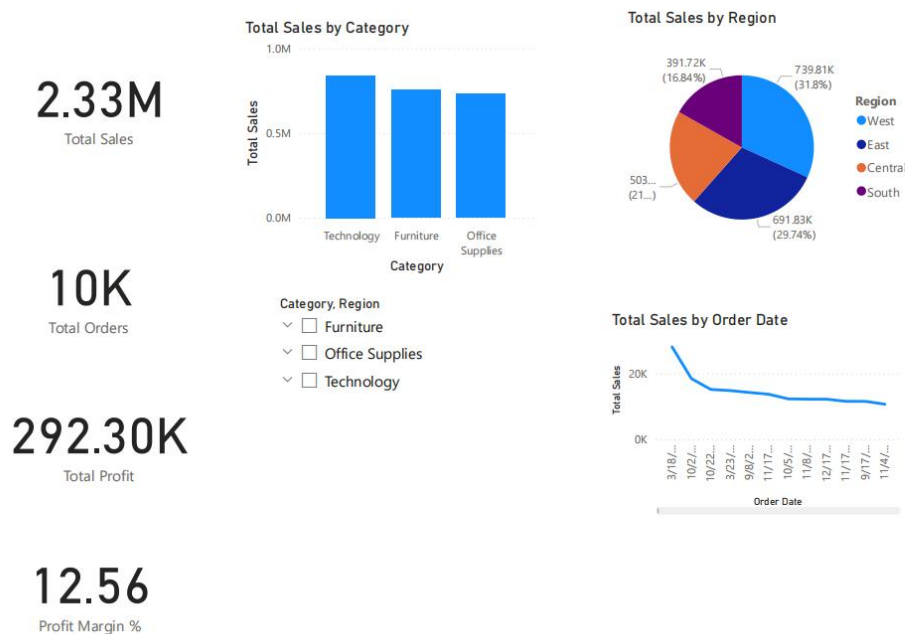
Step 5: Dashboard Formatting

The dashboard was enhanced by:

- Adding appropriate titles for all visuals
- Enabling data labels
- Adjusting font size and alignment

- Including a main title: **“Superstore Sales Dashboard”**

This improved the readability and visual appeal of the dashboard.



Conclusion

In this week, I successfully created an interactive dashboard using Power BI. By applying DAX measures and visualization techniques, I was able to analyze sales performance across categories, regions, and time. This project helped me gain practical knowledge of business intelligence and data visualization.